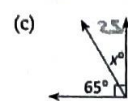
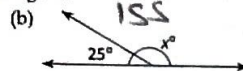
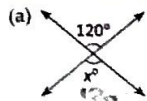


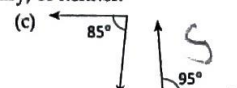
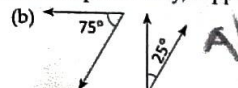
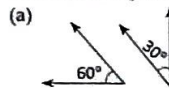
CONCEPT PRACTICE

Exercise 11.1 Related angles

1. Find the value of
- x
- in each figure.



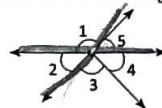
2. Classify each pair of angles as complementary, supplementary, or neither.



3. (a) In a linear pair, if one of the angles is a right angle, what kind of angle will the other one be?
 (b) In a linear pair, if one of the angles is an acute angle, what kind of angle will the other one be?
4. Two supplementary angles are such that one is $\frac{1}{5}$ of the other. Find the measures of both the angles.

5. In the adjoining figure, name the following pairs of angles:

- (a) Vertically opposite angles $\angle 1, \angle 3$ and $\angle 2, \angle 4$
 (b) Linear pairs



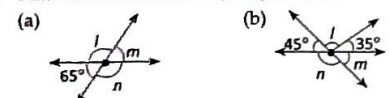
6. Check whether the angles shown in each figure are adjacent to each other. Give reasons.



7. In a pair of supplementary angles, one angle is
- 32°
- less than the other. Find the measures of both the angles.

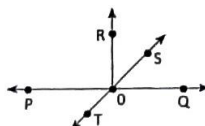
8. The difference between a pair of complementary angles is
- 45°
- . Find the measures of both the angles.

9. Find the measures of
- $\angle l$
- ,
- $\angle m$
- , and
- $\angle n$
- .



10. Identify the following angles in the adjoining figure:

- (a) Adjacent complementary angles
 (b) Equal supplementary angles



- (c) Obtuse vertically opposite angles
 (d) Adjacent angles that do not form a linear pair

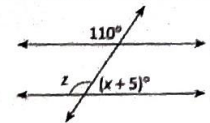
Exercise 11.2 Pairs of lines

1. Look at the adjoining figure and state the property that is used in each of the following statements:

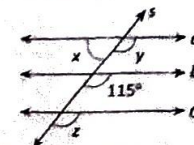
- (a) If $l \parallel m$, then $\angle 1 = \angle 5$.
 (b) If $\angle 4 = \angle 6$, then $l \parallel m$.
 (c) If $\angle 4 + \angle 5 = 180^\circ$, then $l \parallel m$.



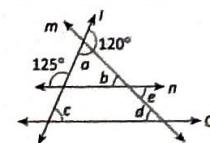
2. If lines
- l
- and
- m
- are parallel in the figure given below, find the measure of
- $\angle z$
- and the value of
- x
- .



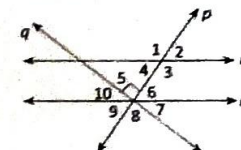
3. Find the values of
- x
- ,
- y
- , and
- z
- when
- $a \parallel b \parallel c$
- and
- s
- is a transversal.



4. Find the measure of angles
- a
- ,
- b
- ,
- c
- ,
- d
- , and
- e
- , if
- $n \parallel o$
- and
- l
- and
- m
- are transversals.



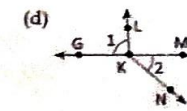
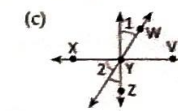
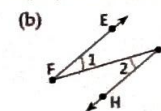
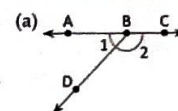
5. In the figure shown below, line
- m
- is parallel to line
- n
- and lines
- p
- and
- q
- are perpendicular to each other. If the measure of
- $\angle 2$
- is
- 45°
- , what is the measure of
- $\angle 10$
- ?



OBJECTIVE QUESTIONS

Choose the correct option for each question.

1. In which of the following figures, angles 1 and 2 are equal?



2. Two angles
- x
- and
- $90^\circ - x$
- are

- (a) a linear pair of angles.
 (b) vertically opposite angles.
 (c) complementary angles.
 (d) supplementary angles.

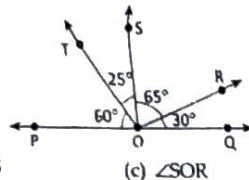
3. The angles
- $y + 15^\circ$
- and
- $190^\circ + y$

- (a) are interior angles on the same side of the transversal.
 (b) form a linear pair.
 (c) are complementary angles.
 (d) none of these.

4. If an angle is
- 30°
- less than two times its complement, what is the measure of the greater angle?

- (a) 40°
 (b) 50°
 (c) 60°
 (d) 70°

5. Which of the following angles is complementary to $\angle SOR$?



(a) $\angle POT$

(b) $\angle TOS$

(c) $\angle SOR$

(d) $\angle$ Both (b) and (c)

Fill in the blanks.

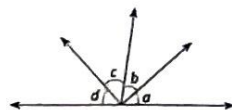
- When two lines p and q meet at a common point, they are called _____.
- The sum of the measures of two _____ angles is 180° .
- When a transversal cuts two lines, _____ pairs of corresponding angles are formed.
- _____ angles are always equal.
- If two lines in a plane never meet at any point, then they are called _____.
- Parallel lines have _____ points in common.
- The supplement of an acute angle is always an _____ angle.
- If two parallel lines are intersected by a transversal, then each pair of alternative angles are _____.
- A _____ intersects two or more than two lines at _____ points.
- If you face the rising sun at an angle of 270° , the direction to your right is _____.

State true or false.

- Two obtuse angles form a pair of complementary angles.
- Vertically opposite angles are always of equal measure.
- Supplementary angles are always obtuse angles.
- A pair of supplementary angles will always have two acute angles.
- Vertically opposite angles form a linear pair.

PUT ON YOUR THINKING CAP!

- A line segment $\overline{XY}$ is parallel to $\overline{PQ}$. $\overline{PQ}$ is perpendicular to $\overline{RS}$. Draw each of the lines. Determine the relation between $\overline{XY}$ and $\overline{RS}$.
- Look at the figure given alongside. What can you conclude about the sum of $\angle a$ and $\angle b$ if $\angle a = \angle b$ and $\angle c = \angle d$? Justify your answer.
- If two angles are supplementary to the same angle, how are the measures of the two angles related?



12

PROPERTIES OF TRIANGLES

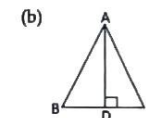
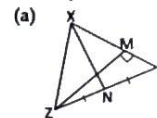
CONCEPT PRACTICE

Exercise 12.1 Types of triangles; Median and altitude of triangles

- Identify the types of triangles in each case.

(a) $AB = 3$ cm, $BC = 3$ cm, $CA = 3$ cm	(b) $AB = 7$ cm, $BC = 5$ cm, $CA = 11$ cm
(c) $AB = 4$ cm, $BC = 5$ cm, $CA = 4$ cm	(d) $\angle A = 40^\circ$, $\angle B = 50^\circ$, $\angle C = 90^\circ$
(e) $\angle A = 60^\circ$, $\angle B = 60^\circ$, $\angle C = 60^\circ$	(f) $\angle A = 60^\circ$, $\angle B = 70^\circ$, $\angle C = 50^\circ$
(g) $AB = PR = 7$ cm, $\angle A = 90^\circ$	

- Identify the medians and altitudes of the following triangles.



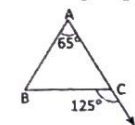
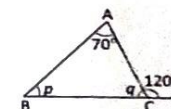
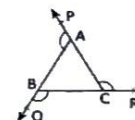
In which cases do the medians and altitudes overlap?

- Draw three triangles in which the altitudes lie inside, on the side, and outside the triangles.

Exercise 12.2 Properties of triangles

- If the vertex angle of an isosceles triangle is of measure 80° , find the measures of base angles.
- If the angles of a triangle are in the ratio of 2:5:3, find the measures of the angles.
- Find the value of z if the three angles of a triangle are $8z$, $5z$, and $5z$.
- Look at the figure shown alongside and fill in the blanks:

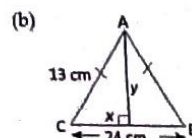
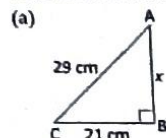
(a) $\angle ACR = _____ + _____$
(b) $\angle PAB = _____ + _____$
(c) $\angle QBC = _____ + _____$
- In $\triangle PQR$, the measures of $\angle P$ and $\angle Q$ are in the ratio of 3:7. If the exterior angle $\angle PRM$ of the $\triangle PQR$ is of measure 100° , find the measures of all the angles of the triangle.
- Find the values of p and q in the figure given below.
- Find the measures of $\angle ABC$ and $\angle ACB$ in the figure given below.



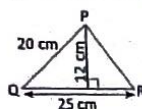
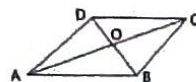
8. One side of a triangle is extended and an exterior angle of measure 130° is formed. If the measures of interior opposite angles are in the ratio 6:7, find the measures of all the angles of the triangle.

Exercise 12.3 Sum of the lengths of two sides of a triangle, right-angled triangles, and the Pythagoras' theorem

- Which of the following measures can form a triangle?
 - 6 cm, 8 cm, and 10 cm
 - 5 cm, 10 cm, and 15 cm
 - 8 cm, 6 cm, and 13 cm
 - 7 cm, 7 cm, and 7 cm
- Check if the following set of numbers form Pythagorean triplets.
 - 1.4 cm, 2.3 cm, and 3.4 cm
 - 7 cm, 24 cm, and 25 cm
 - 8 cm, 14 cm, and 16 cm
 - 9 cm, 40 cm, and 41 cm
- Find the unknown lengths.



- Look at the quadrilateral ABCD given alongside. Now check which of the following statements are true?
 - $DC + CB > BD$
 - $AB + 2BC > AC$
 - $AB + BC + CD + AD > 2AC$
 - $BD + AB < AD$
- Look at the triangle PQR given alongside. Find
 - the length of the side PR and
 - the area of the triangle PQR.



- A ladder leans against a wall of a building. The length of the ladder is 5 m and its base is 3 m away from the building. Find the height of the wall.
- A playground is of length 144 m and breadth 17 m. If the coach asked the players to run from one corner to the other corner diagonally across the playground, what distance did each player run?
- The heights of two towers are 15 m and 10 m. If the distance between the top of the two towers is 13 m, what is the distance between the bases of the two towers.

OBJECTIVE QUESTIONS

Choose the correct option for each question.

- What is the relationship between the two acute angles of a right-angled triangle?
 - They are always equal.
 - They are complementary.
 - They are supplementary.
 - They are never equal.
- $\triangle PQR$ is an isosceles triangle. If the measure of $\angle Q$ is 76° and the measures of $\angle P$ and $\angle R$ are equal, what is the measure of $\angle P$?
 - 42°
 - 49°
 - 52°
 - 69°
- Which of the following is possible in a triangle?
 - Two right angles
 - Two acute angles
 - Two obtuse angles
 - Two angles forming a linear pair

- The hypotenuse and a side of a right triangle are of lengths 10 cm and 6 cm respectively. The length of its third side is equal to _____.
 - 11 cm
 - 8 cm
 - 9 cm
 - 6 cm
- What is the maximum number of altitudes in a triangle?
 - 1
 - 2
 - 3
 - 4

Fill in the blanks.

- The two sides of a triangle are of lengths 11 cm and 13 cm. The length of the third side cannot be _____ than or _____ to the sum of the two sides.
- If the measure of the base angle of an isosceles triangle is 40° , the vertex angle will be of measure _____.
- If the exterior angle of an isosceles triangle at the vertex (non-base side) is of measure 140° , the measure of each of the equal angles is _____.
- The minimum number of acute angles in a triangle is _____.
- In a right-angled triangle if one acute angle is 55° , the third angle is of measure _____.

State true or false.

- There is only one altitude in a triangle from the same vertex.
- The median of a triangle can lie in the exterior region of the triangle.
- A triangle can have two acute angles.
- The measures of the three angles of a triangle can be 40° , 45° , and 60° .
- A median will always lie in the interior region of a triangle.



PUT ON YOUR THINKING CAP!

- Babita says that in an equilateral triangle, medians and altitudes are the same. Is she correct? Justify your answer.
- If a square has been drawn on each side of a triangle, how can you tell whether the triangle is obtuse, acute, or right angled?
- Look at the figure given alongside and prove that $A + B + C + D + E + F = 360^\circ$.

